Chapter 2

DETECTION OF ALKALOIDS IN TLC

For the detection of alkaloids many different methods have been described, ranging from non-selective to highly selective ones. Most of them were already in use in combination with paper chromatography.

The non-selective methods, which are used for the detection of all kind of organic compounds in TLC, such as quenching of UV light on fluorescent plates, iodine vapour or iodine spray reagents and concentrated sulphuric acid, are usually fairly sensitive, allowing a detection limit of less than 1 µg. However, because of a lack of specificity of these reagents other methods are usually preferred for the detection of alkaloids. Methods by means of which alkaloids can be selectively detected are particularly important. Selective and specific alkaloids reagents are various modifications of Dragendorff's reagent and potassium iodoplatinate. Both reagents react with tertiary and quaternary nitrogen atoms.

2.1. DRAGENDORFF'S REAGENT (REAGENT 39a-39g)

The modifications of this reagent were originally described for the detection of alkaloids on paper chromatograms. The first spray reagent described for this purpose were the modifications proposed by Munier and Macheboeuf[1] and Munier[2]. Both modifications are aqueous solutions, the difference being the acid used to dissolve the bismuth salt (acetic and tartaric acid, respectively). The sensitivity of the reagent for alkaloids is in the range 0.01-1 µg.

Thies and Reuther[3] developed a modification using a solution in acetic acid and ethyl acetate instead of water. Because the ethyl acetate evaporates faster than water, less diffusion of the spots will occur, resulting in sharper edges of the spots. However, the sensitivity is lower than that of the two already mentioned modifications[4].

Vágújfalvi[4] modified the method of Thies and Reuther by spraying the chromatogram with 0.05 *N* sulphuric acid after spraying with Dragendorff's reagent. The sensitivity increased ten-fold in paper chromatography (PC). The author assumed that the complex formation with the ethyl acetate-containing reagent is different from that with the aqueous reagent.

Later this method was modified for TLC by Vágújfalvi[5]: 10% sulphuric acid was used instead of 0.05 *N* sulphuric acid to increase the sensitivity of the reaction.

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According to the same author, maximum intensity in colour is observed after 15-60 min; however, exposure to strong light for 15-30 min makes the colour decrease. Further, it was stated that a four-fold more dilute reagent than used in PC was more sensitive for alkaloid detection in TLC.

Trabert[6] described a Dragendorff reagent dissolved in diethyl ether-methanol. In this form the reagent is much more sensitive in detecting alkaloids in ethereal solutions and on paper chromatograms. Robles[7] stated that hydrochloric acid is more suitable than acetic acid in preparing Dragendorff's reagent, because too high a pH leads to faster fading of the spots. Dilution of the reagent should be done with 0.1 *N* hydrochloric acid instead of water, otherwise a cloudy solution is obtained. The same is observed if the amount of potassium iodide is too small.

If alkaloids are to be detected on a buffered cellulose layer, it is necessary to add a non-volatile acid. Citric acid can be used with good results. If bismuth nitrate is used instead of bismuth subnitrate, the colour of the alkaloid spots is more intense[7].

Primary and secondary amines are not (or only at high concentrations) detectable with Dragendorff's reagent. However, treatment of such compounds with dimethyl sulphate, giving quaternary nitrogen atoms, enables detection with Dragendorff's reagent to be achieved[8].

Roper et al.[9] reported on the sensitivity and the stability of Dragendorff's reagent prepared according to Munier and Macheboeuf. They found that neither the reagent concentrate nor the diluted spray reagent showed any decrease in alkaloid sensitivity under normal laboratory storage conditions for at least 6 months. There seemed to be no reason for storing either the reagent concentrate or the diluted spray reagent in amber-glass bottles or under refrigeration. However, the best results were obtained when the spray reagent was prepared from a reagent concentrate at least 2 days after its preparation, and further that spray reagents made from such concentrates should not be used before 6 days after dilution.

To increase the sensitivity of the reaction with Munier's or Munier and Macheboeuf's modification of Dragendorff's reagent, Fike[10], Puech et al.[11] and McLean and Jewers[12] sprayed afterwards with 10% sodium nitrite solution. By this treatment the background of the spots is decolorized from yellow to white, improving the contrast of the spots. For several groups of alkaloids the sensitivity is increased in this way, the sensitivity being in the range 0.01 - 0.1 μg[13].

Various non-alkaloidal compounds also react with Dragendorff's reagent[8,13-15]. In Table 2.1 examples of such compounds are summarized. Other compounds which are reported to precipitate with Dragendorff's reagent are proteins, including albuminous substances, peptones, ptomaines[15] and choline[16]. However, amino acids do not give any precipitate with Dragendorff's reagent[17].

Except for choline, the compounds mentioned have not been tested in TLC for a

TABLE 2.1

NON-ALKALOIDAL COMPOUNDS WITH POSITIVE DRAGENDORFF REACTIONS

Positive reaction: minimum detectable amount 10 µg or less[13-15]

Type	Compounds with positive reactions
α-Pyrone	Kawain, yangonin, desmethoxyyangonin, dihydrokawain, methysticin, dihydromethysticin and some synthetic products
Benz-α-pyrones	Coumarin, scopoletin, bishydroxycoumarin, ethylbis(4-hydroxy-coumarinyl) acetate
Psoralens	bergapten, xanthotoxin, imperatorin, isopimpinellin
γ-Pyrones	Maltol, kojic acid, kojic acid methyl ether
Furochromones	Khellin
Chalcones	Chalcone, β-methoxychalcone, *cis*- and *trans*-1,4-diphenyl-2-butene-1,4-dione, cardenolides, steroids and triterpenes, digitoxin, ouabain, sitosterol, stigmasterol, cholesterol, quillaic acid, β-amyrin, lupenone, β-glycyrrhetinic acid, oleanolic acid. Friedelanol and cholestane are observed after spraying with sodium nitrite
Miscellaenous compounds	Acrolein, amylcinnamaldehyde, cinnamaldehyde, eugenol, hydroxy-citronellal, menadione, menthyl salicylate, ninhydrin, phenyl salicylate, salicylic acid, benzyl acetate, camphor, eucalyptol, methyl salicylate, piperonal, 2-phenylethanol, cinnamyl alcohol, choline, phosphatidylcholine. Geraniol, resorcinol, orcinol, anethol and thymol give a positive reaction on spraying with sodium nitrite. Further, acetylsalicylic acid, cinnamic acid, guaiacol, hydroquinone, phloroglucinol, quinone, terpineol and vanillin give a positive reaction after 24 h.
Polyethylene glycols	

possible positive or negative Dragendorff reaction. Farnsworth[17] surveyed the false-positive and false-negative reactions of alkaloids. According to Farnsworth et al.[15], the minimum structural features for non-alkaloids to give a positive Dragendorff reaction are conjugated carbonyl (ketone or aldehyde) or lactone functions. Anderson et al.[13] observed that the sensitivity for such compounds increased if the plate was sprayed with sulphuric acid or sodium nitrite after spraying with Dragendorff's reagent. The authors concluded that the minimum structural requirement for a positive Dragendorff reaction is a hydroxyl group and an isolated double bond.

2.2. POTASSIUM IODOPLATINATE (REAGENTS 64a-64f)

This reagent also exists in various modifications. The sensitivity is about the same as for Dragendorff's reagent (0.01 - 0.1 µg), but potassium iodoplatinate reagent has several advantages.

It is non-destructive and the alkaloids can be recovered after application of the spray reagent. Holdstock and Stevens[18] used the following method for the recovery of the alkaloids. After drying the plate the spot material was scraped off the plate and transferred to a test-tube. A few drops of a saturated sodium sulphite solution were added, followed by 1 ml of 0.5 *N* sulphuric acid. If necessary the mixture was heated to decolorize the sample. The aqueous solution was then saturated with sodium chloride, followed by basification with strong ammonia. Subsequently the alkaloid bases could be extracted with diethyl ether, chloroform or butanol-chloroform (1:9). A recovery of 50-60% was found for tertiary amines and 70-80% for primary and secondary amines. For application in PC, see also Goldbaum and Kazyak[19].

Another advantage of potassium iodoplatinate reagent is that different colours are obtained with different alkaloids, the colours varying from brown through violet to blue. A blue-violet background colour due to starch present can be decolorized by spraying with sodium hydrogen sulphite solution.

Formamide impregnation of TLC plates can also interfere with the alkaloid detection, but by spraying the plate with a 0.25% solution of sodium nitrite in 0.5% hydrochloric acid the formamide will be converted into formic acid, which does not interfere with the alkaloid detection.

Diethylamine also interferes with alkaloid detection when using Dragendorff's reagent or iodoplatinate reagent. Even after drying the plate at elevated temperature the diethylamine does not disappear completely, resulting in a dark background after spraying with one of the reagents mentioned, with a consequent decrease in detection sensitivity[11].

2.3. OTHER REAGENTS

In addition to the two widely used reagents mentioned above, various other reagents have been described for the detection of alkaloids.

Marquis reagent (formaldehyde in sulphuric acid, no. 53)[20,30], ammonium vanadate in nitric or sulphuric acid (nos. 3, 91 and 103)[21,22,29], Fröhde's reagent (sulphomolybdic acid, no. 90)[21,29] and phosphomolybdic acid in nitric acid (no. 79)[21] have been used in the detection of alkaloids. Two reagents which may be mentioned separately because of their usefulness in the selective detection of several groups of alkaloids are iron(III) chloride in perchloric acid (nos. 46a-46c) and cerium(IV) sulphate in sulphuric acid or orthophosphoric acid (nos. 12 - 14c).

Rücker and Taha[23] described the use of π-acceptors for the detection of alkaloids. Several reagents were used. In Table 2.2 the colours observed with these reagents are summarized for some alkaloids. TCNQ reagent (no. 95) was found to be the most sensitive and to have the widest applicability. The sensitivity was in the range of 0.5-10 μg. Vinson and co-workers[24,25] described the use of TCBI reagent (no. 94) for the detection of drugs, including a number of alkaloids. Various colours are obtained for the test compounds (see Table 2.3), the detection limit being in the range 0.05-0.25 μg.

Grant[26] described the use of aqueous cobalt thiocyanate solution in the detection of various drugs, including a series of alkaloids. Several alkaloids give positive reactions. Sometimes derivatization may lead to a positive reaction, e.g., acetylation of morphine. Ranieri and McLaughlin[27] described the use of fluorescamine for the detection of primary and secondary amines. They proposed the following procedure. The TLC plate with the alkaloid extract is first sprayed with the fluorescamine solution (no. 50) (detection of primary and secondary amines), then with dansyl chloride reagent (no. 28) (detection of phenols, imidazoles and the fluorescamine conjugates of the secondary amines are converted into dansyl conjugates) and finally with iodoplatinate (detection of tertiary amines). Menn and McBain[28] described a detection method specifically for cholinesterase inhibitors. First the plate is sprayed with human plasma, and after 30 min with a mixture of 1 part of 0.6% bromothymol in 0.1 *N* sodium hydroxide solution and 15 parts of 1% aqueous acetylcholine chloride solution. The inhibiting substances show a blue colour against a yellow background. This method is applicable only on cellulose plates and does not work on silica gel or Florisil plates.

For each group of alkaloids the appropriate detection methods will be discussed in more detail. A list of spray reagents and their preparation are given in the Appendix.

TABLE 2.2

COLOURS OF TLC SPOTS OF ALKALOIDS WITH π-ACCEPTORS[23]

Colours: Bl = blue; Br = brown; Gn = green; Go = gold; Gy = grey; nc = no colour; Or = orange; R = red; V = violet; W = white; Y = yellow. White background unless otherwise stated.

Alkaloid	TCNQ* (no.95)	TNF (no. 100b)		TetNF (no.97b) after heating	DDQ** (no.29)	DNFB (no.38)
		In the cold	After heating			
Atropine	Y-W	nc	V	Gn-Gy***	Or	Y
Scopolamine	Gn-Y	nc	Gy	Gy***	nc	Y
Homatropine	Go	nc	V	V-Gy***	Or	Y
Homatropine·MeBr	Go	nc	R	V***	nc	Y-W
Atropine·$MeNO_3$	Y-Or	nc	R	V***	nc	Y-W
Tropine	Go	nc	V-Gy	V***	Gn	Y-W
Pilocarpine	Go	nc	Gy	V-Gy***	V	Y
Ephedrine	Gn-Y	nc	Gy	Faint***	nc	Y-Or
Cocaine	Go	nc	V-Gy	V***	nc	Or
Morphine	Bl	Or-Br	Or	Or	Gn	Or
Codeine	Y-Go	Or-Br	Y	Gy	V-Br	Or
Papaverine	Y-Go	Or	Y	Br	V	Or
Quinine	Y-Go	Y	V-Br	V	V	Or
Brucine	Go	V	V	Gy	V	Or
Strychnine	Y	Y	Y	Or	Or	Or
Veratrine	Go	Or	Or	Gy	Or	Or
Reserpine	Gn	Gy	Gy	Or	Gn	Or-R
Ergotamine	Y-Gn	Gy-Br	Gy-Br	Gy	Br	Or-R

*Colours given were on a pale bluish green background.
**Pale violet background.
***No colour in the cold.

TABLE 2.3

COLOURS OF VARIOUS DRUGS SPRAYED WITH TCBI (No.94)[24]

Drug	Colour	Drug	Colour
Acetophenazine	Yellow-green	Mephenytoin	Blue
Amobarbital	Blue	Mescaline	Pink-brown
Amphetamine	Grey-purple	Methadone	Blue-green
Anileridine	Brown-green	Methamphetamine	Green
Apomorphine	Blue-green	Methapyrilene	Brown-green
Atropine	Blue-green	Methyldopa	Brown
Benzocaine	Orange-brown	Methylphenidate	Grey-green
Bufotenine	Brown-green	Nicotine	Grey-purple
Chlordiazepoxide	Green	Nortryptiline	Brown
Chloroquine	Green	Oxazepam	Blue
Chlorothiazide	Grey-brown	Pentazocine	Grey-green
Chlorpromazine	Grey-purple	Perphenazine	Grey-purple
Cocaine	Green	Phendimetrazine	Grey
Codeine	Brown-green	Phenmetrazine	Yellow-green
Dextromethorphan	Grey	Phenobarbital	Blue
Dicyclomine	Green	Phentermine	Green
Diethylpropion	Grey-brown	Phenylephrine	Green
Dimethyltryptamine	Brown-green	Procaine	Brown-green
Diphenylhydantoin	Blue-grey	Promazine	Purple
Diphenhydramine	Blue-green	Promethazine	Purple
Doxylamine	Brown	Quinine	Grey-green
Ephedrine	Grey-green	Quinidine	Green
Epinephrine	Brown	Reserpine	Brown
Ethinamate	Orange-brown	Secobarbital	Blue
Glutethimide	Grey-green	Strychnine	Green
Heptabarbital	Blue	Sulfamerazine	Grey
Heroin	Yellow-green	Sulfathiazole	Brown-blue
Hydrochlorothiazide	Grey-purple	Tetrahydrocannabinol (Δ^9)	Red-brown
Ibogaine	Brown-green	Thiopropazate	Blue
Lidocaine	Green	Thioridazine	Grey-brown
Lysergic acid diethylamide	Grey-brown	Trifluoperazine	Blue-purple
Meperidine	Grey-green	Trimeprazine	Blue-grey
Mephentermine	Green		

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